

Submission & Reviewer Matching

Weekly update from Qiang

08/13/2019

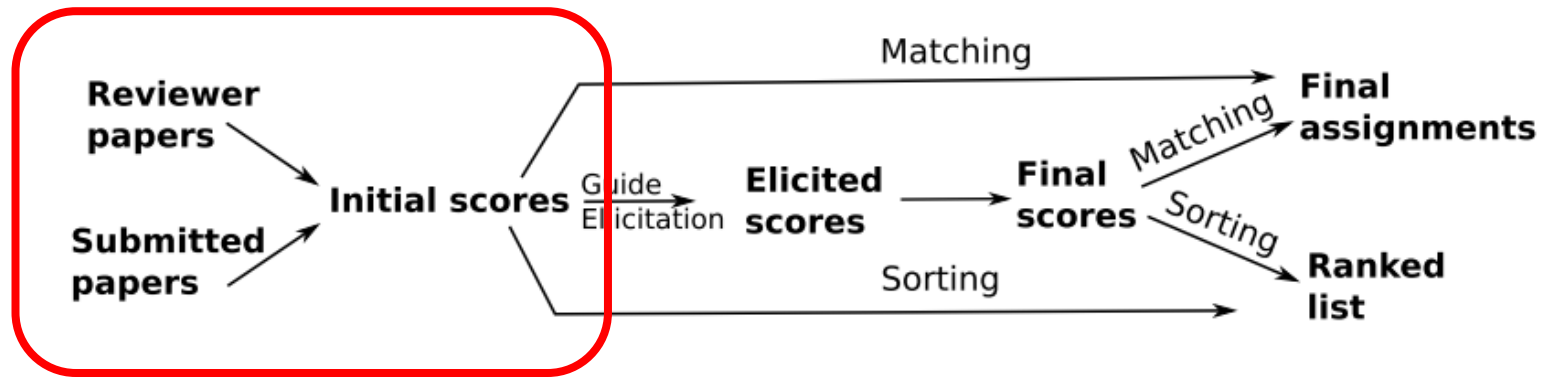
Update

1. Initial try of TFIDF
2. Evaluation metric
3. Word embeddings
4. Topic weights
5. Re-organized the code
6. Summary of previous results
7. Initial try of ELMo
8. Retry ELMo
9. Proposal of next steps

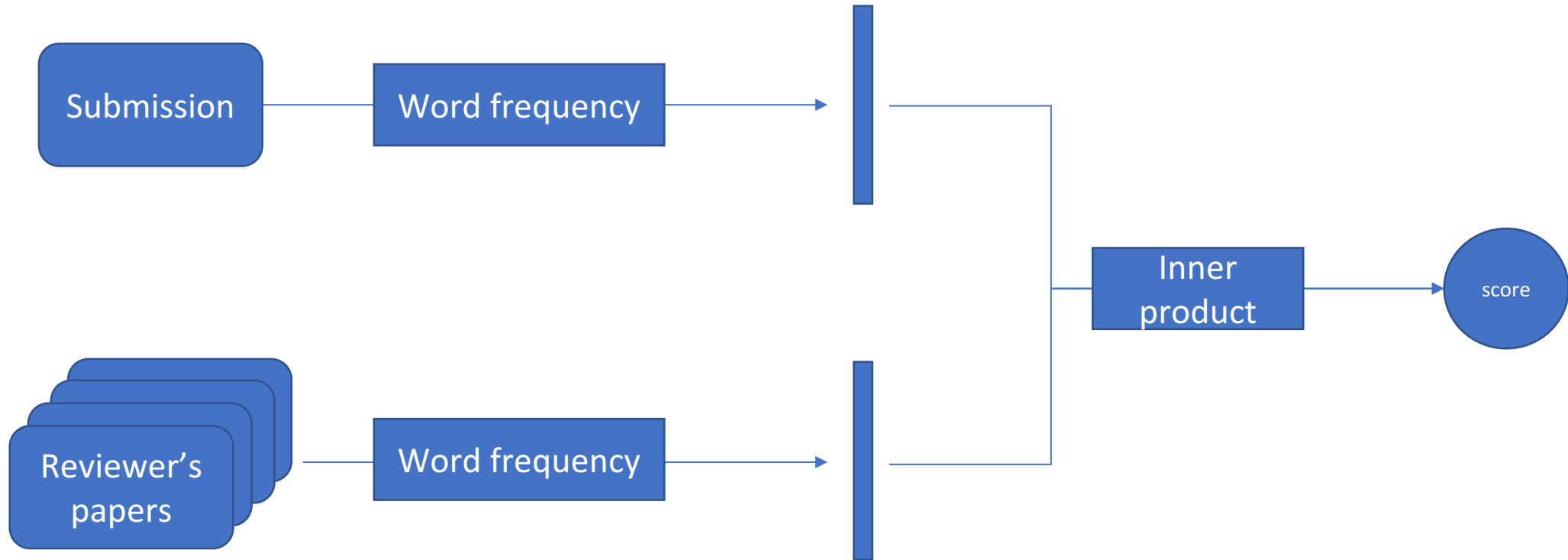
Summary of the conclusions we have so far

- Contextualized embeddings are not necessarily improving for this problem
- Solution to this task heavily relies on extracting relevant keywords
- In-domain embeddings are better than off-the-shelf embeddings
- Evaluation metrics for ranking problems are suitable here.
 - F1@K, Acc@K, MRR
 - NDCG, Spearman's rho, mean average precision
- Computationally:
 - hundreds of reviewers and hundreds of papers should be solved within minutes on a single desktop
 - a lot of computations can be done and saved before the assignment starts (e.g., vectorization of reviewers)
 - Highly parallelizable: similarity scores can be computed independently for each submission

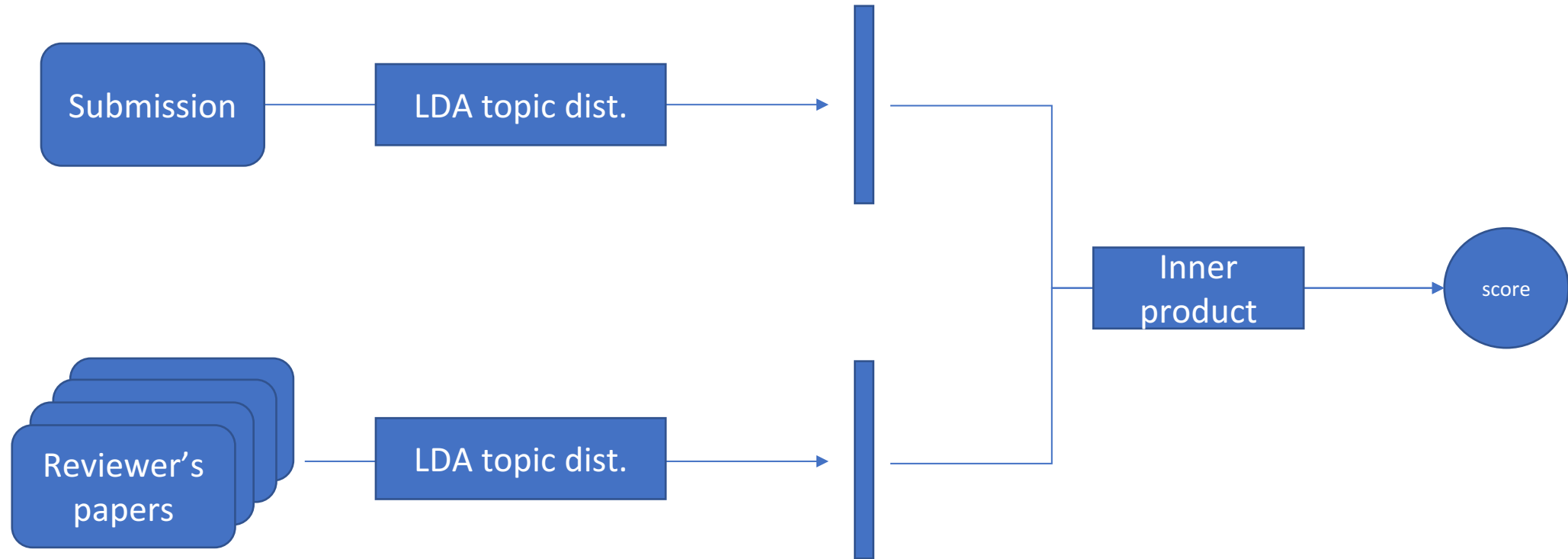
High-level workflow



What is TPMS doing? Method 1

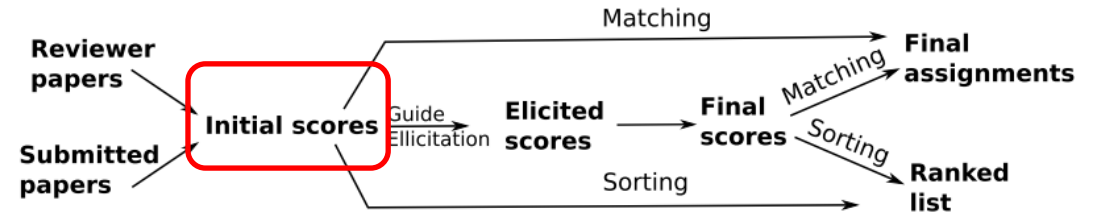


What is TPMS doing? Method 2



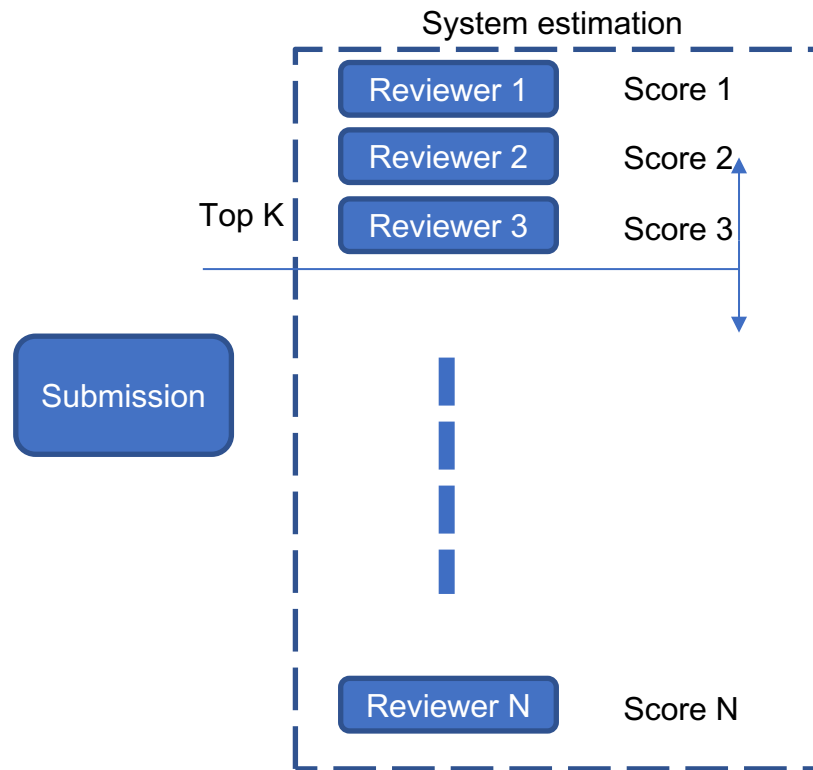
LDA can group different words into the same topic, so it's conceptually still "word frequency" but with sparsity alleviated.

Proposal



- Try two baselines and understand where we are:
 - Compare with TPMS (since the method is straightforward, we can re-implement it if there's no pkg available). I think we should be better than TPMS at least in terms of the "initial scores."
 - Try existing citation recommendation works (to confirm that citation network is useful for us)
- Construct our own reviewer matching dataset (see next)

Difficulty in comparing with previous works



System 1
prediction

Human
labels

Expert

Non-expert

Expert

Expert

Expert

Non-expert

Non-expert

Unlabeled

Non-expert

Unlabeled

Non-expert

Unlabeled

Non-expert

Unlabeled

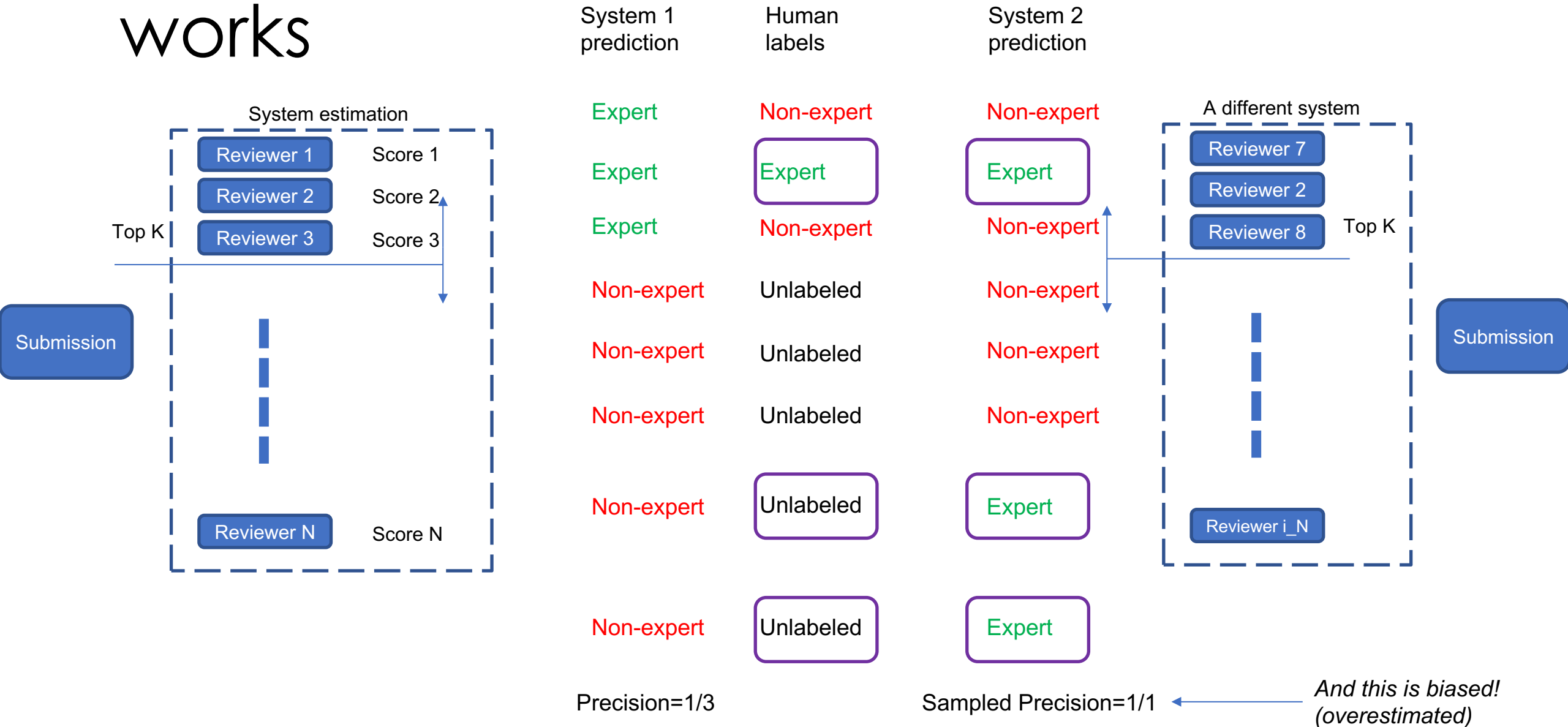
Non-expert

Unlabeled

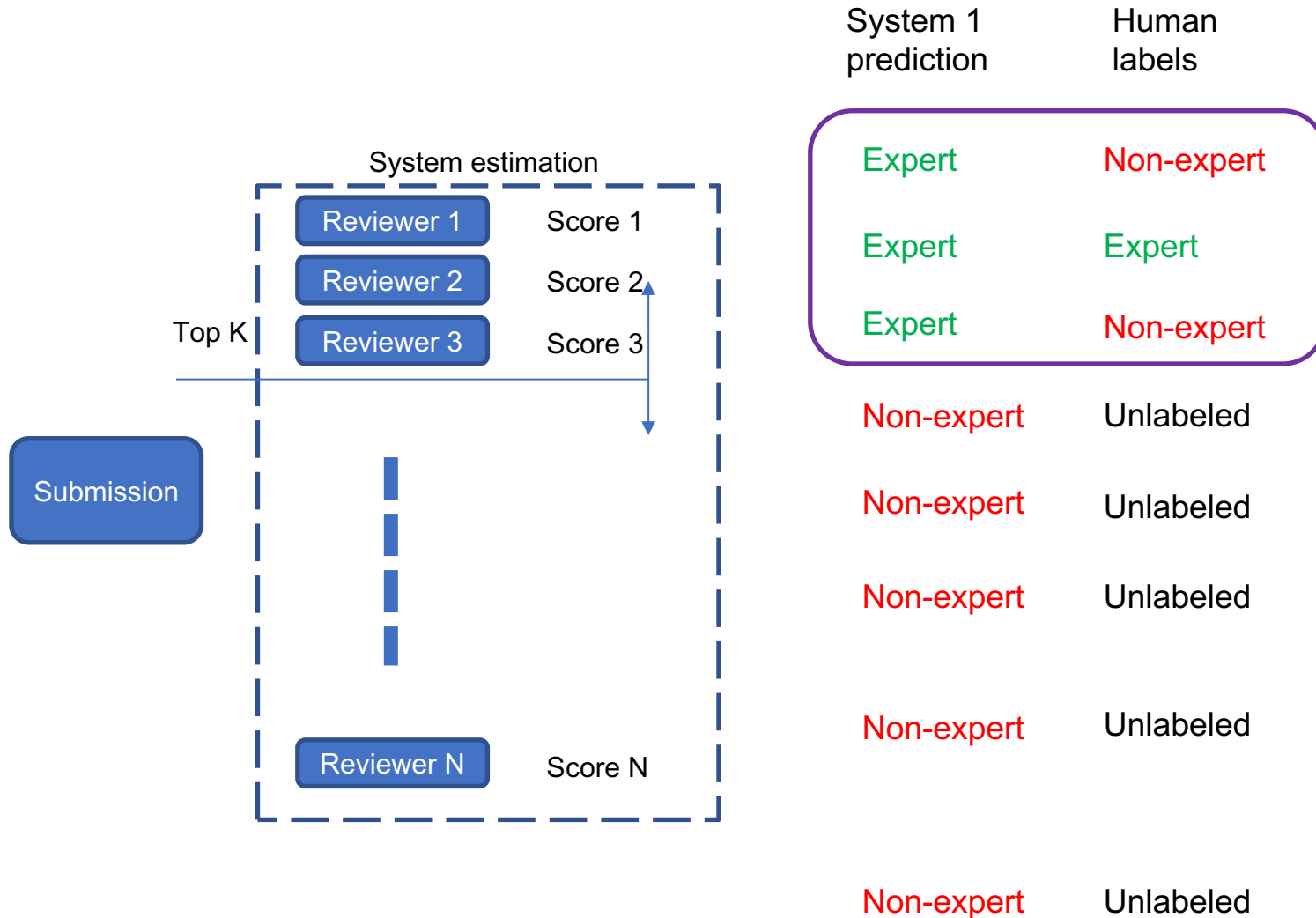
Compare with human labels and we can get an estimate of how good our system is.

So far so good.

Difficulty in comparing with previous works

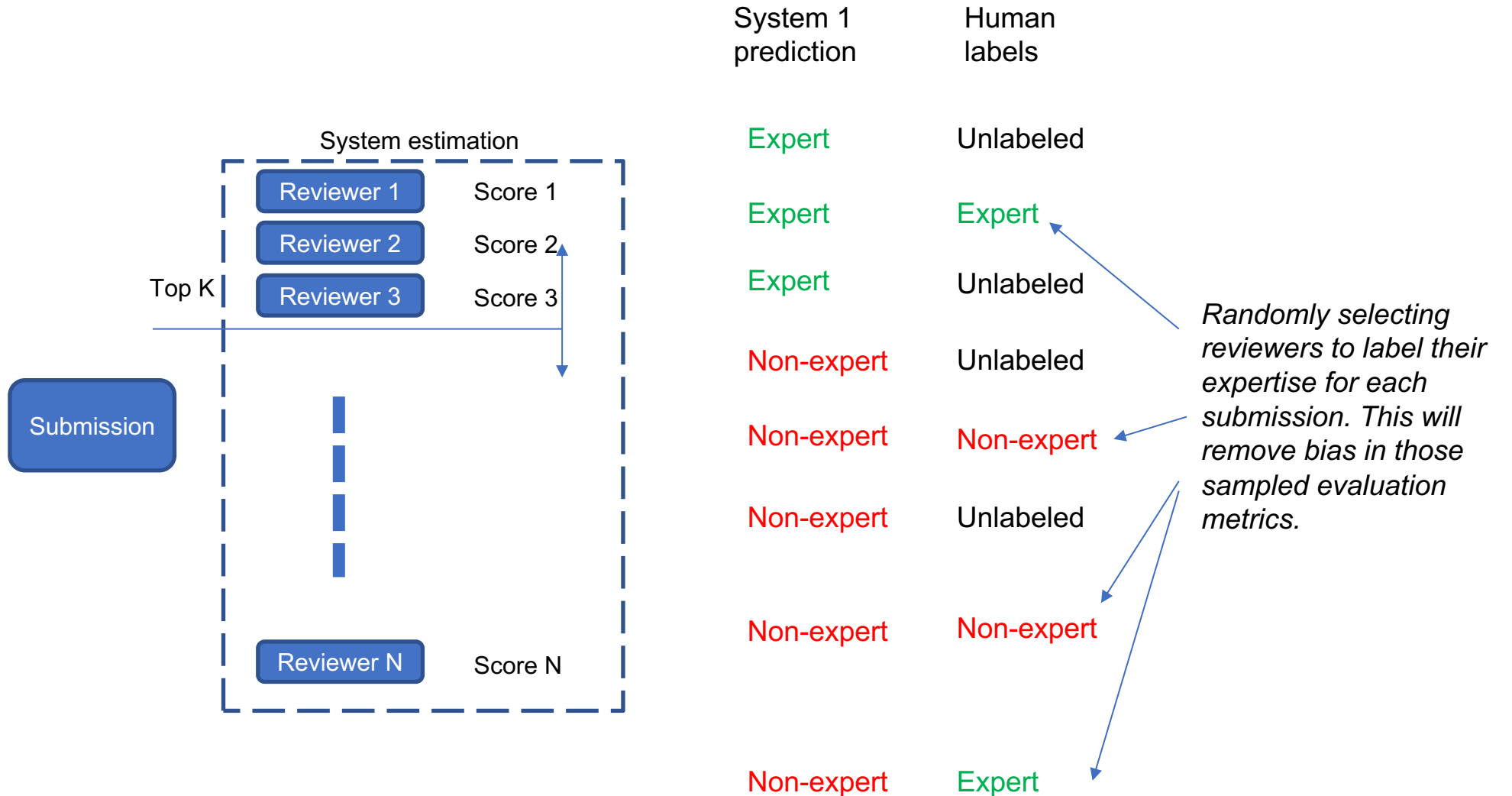


Construct our own reviewer matching dataset: Type 1

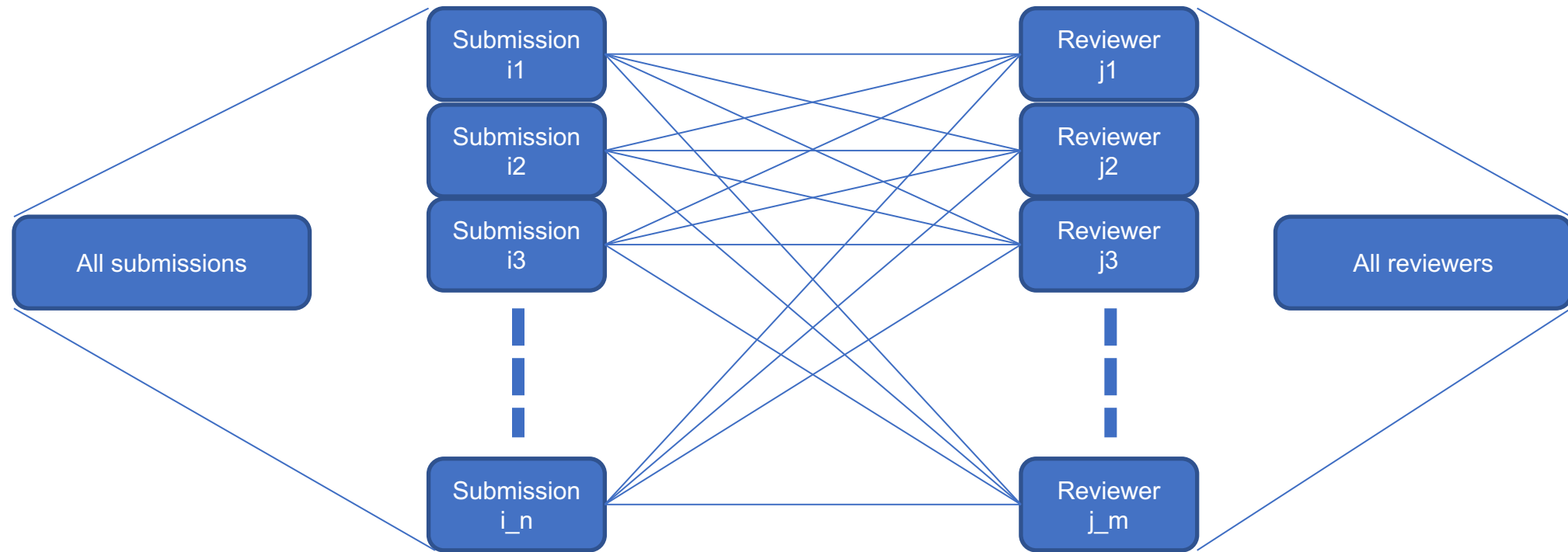


Just like before so that we can get the most reliable performance for our own system.

Construct our own reviewer matching dataset: Type 2



Construct our own reviewer matching dataset: Type 3

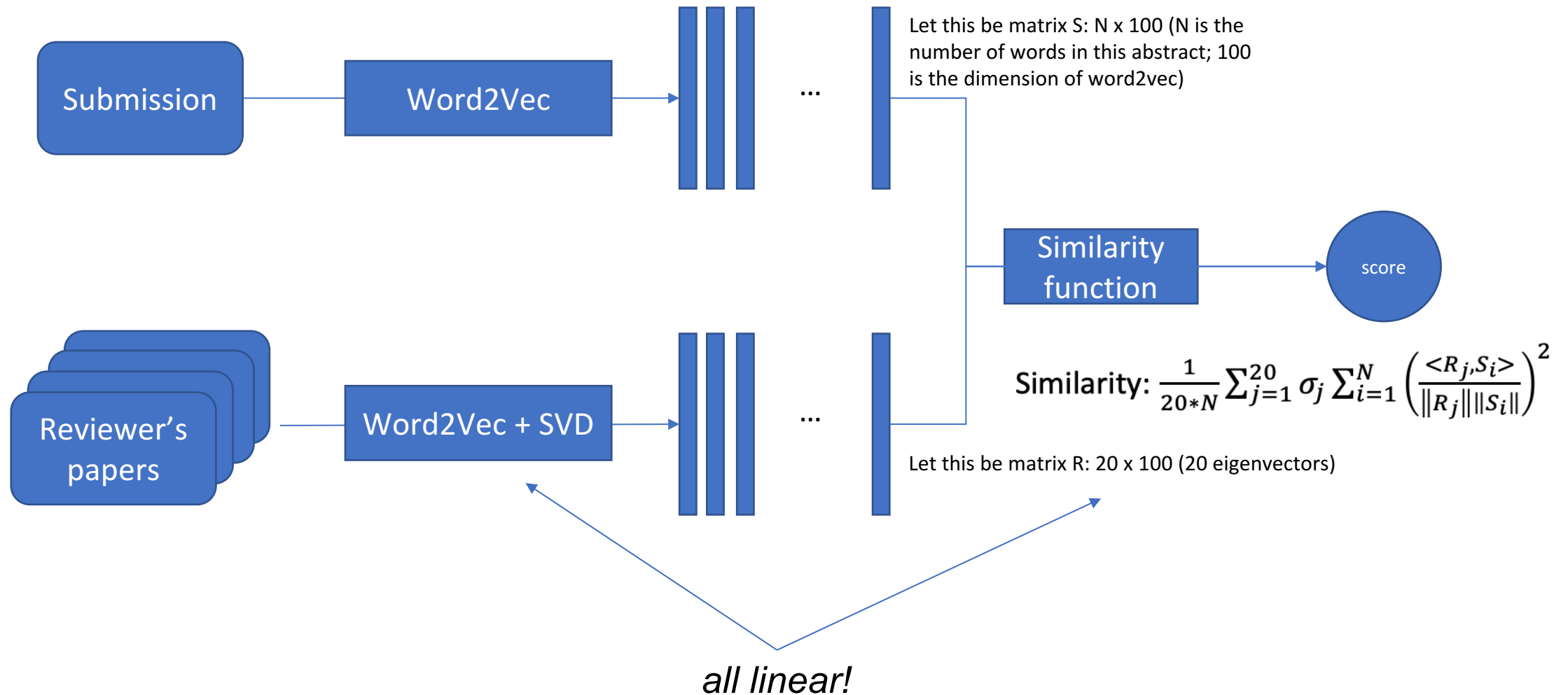


Randomly select n submissions and m reviewers. Ask ALL reviewers to label their expertise on ALL submissions. We have to select a small n and m , but this will be the first fully annotated dataset, although this is small.

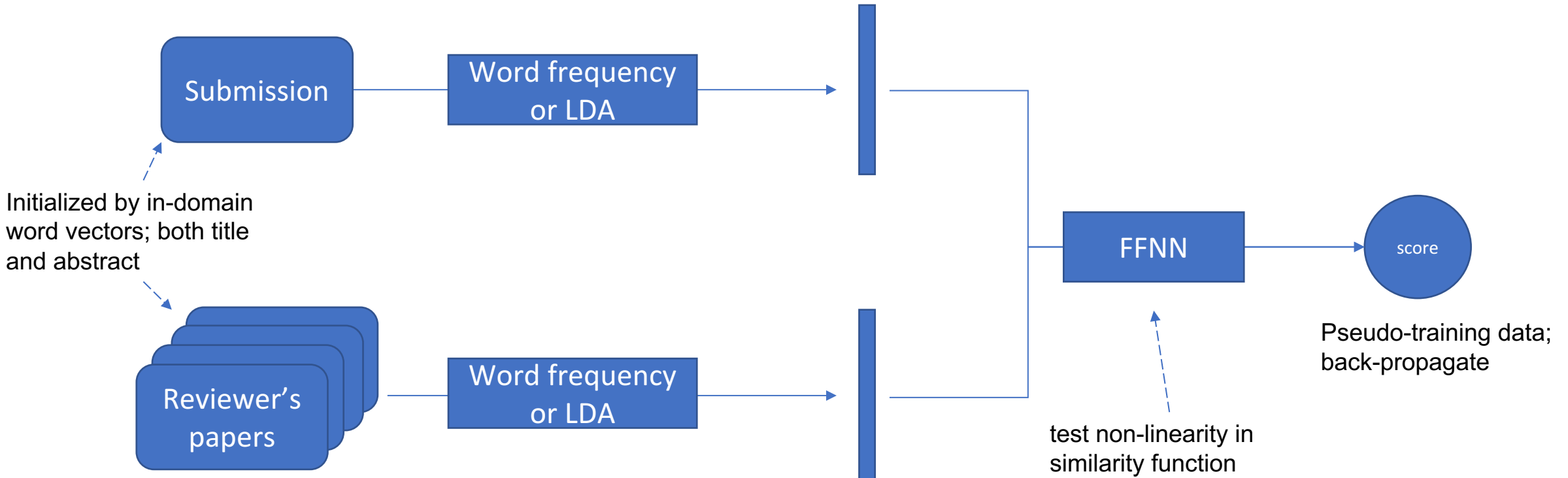
Proposal

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- Construct our own reviewer matching dataset
- Construct pseudo-reviewer datasets:
 - Dataset A: from a citation network. For example, the author whose work is cited most in paper X should be a good reviewer for paper X.
 - Dataset B: from one's own papers. For example, if one authored paper $\{1, 2, \dots, N\}$, then a pseudo-author for $\{1, 2, \dots, N\} \setminus \{i\}$ should be a good reviewer for paper i .
 - ...
- We can then use these pseudo-reviewer datasets to do supervised training (as I show next).

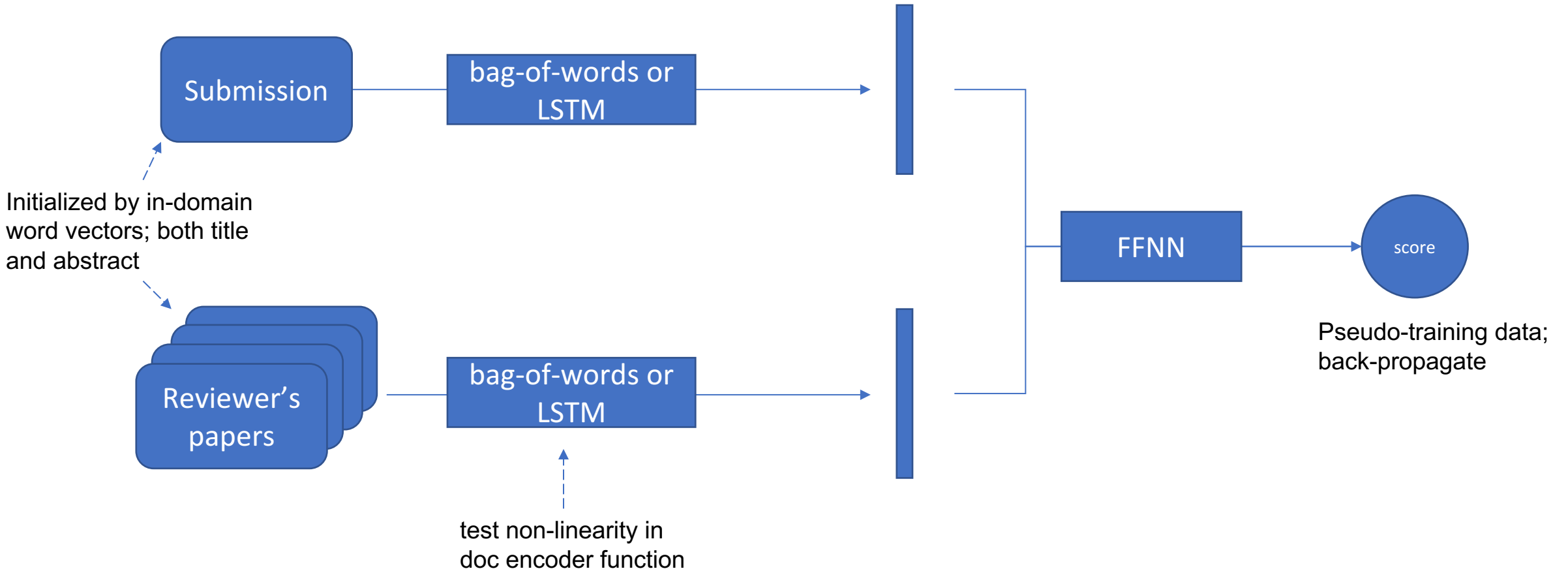
Current system



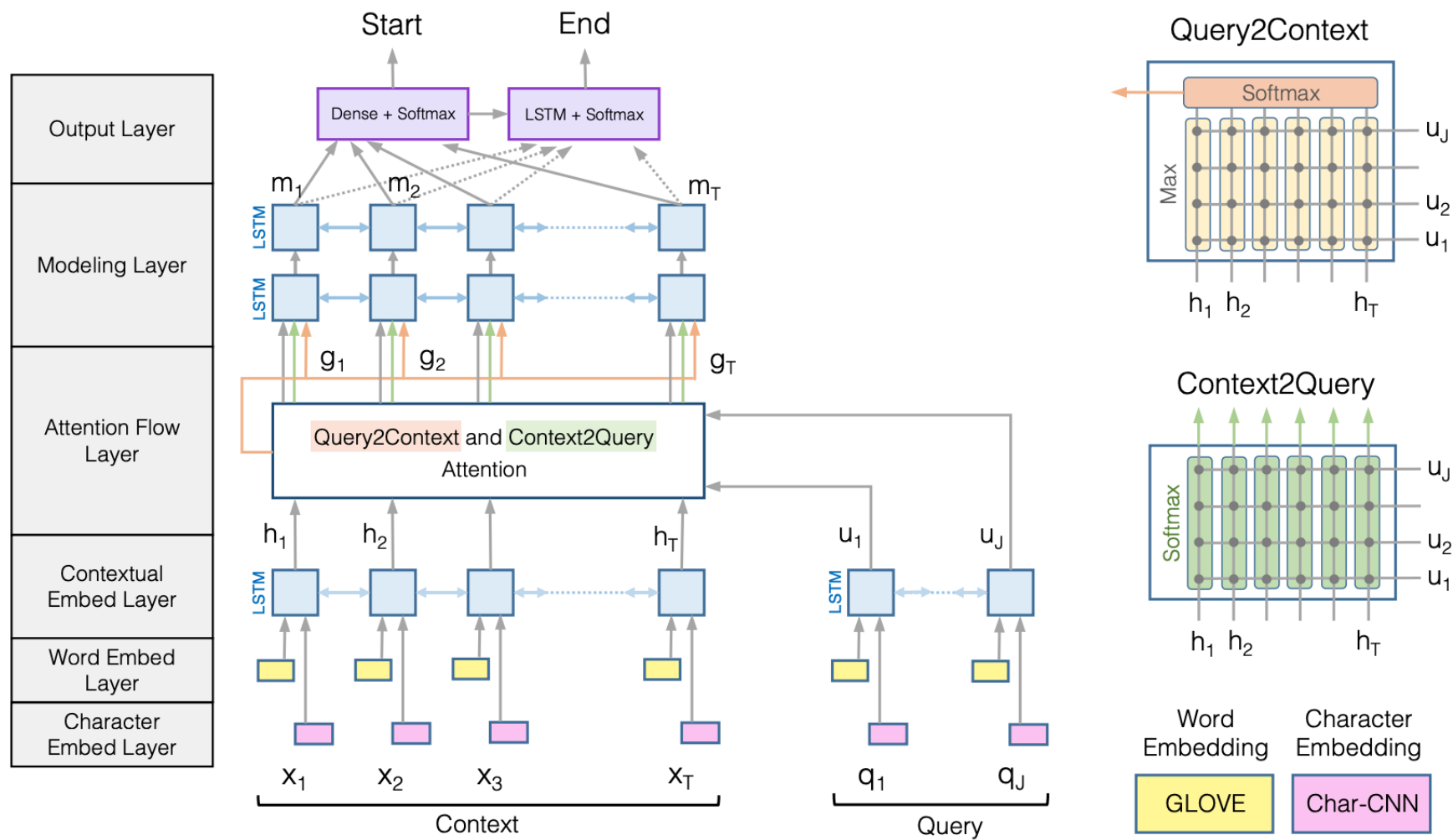
Proposal: model 0



Proposal: model 1



biDAF: Bidirectional Attention Flow



Proposal: model 2

